For Examiner's Use

2 **(a)** State Newton's law of gravitation.

.....

.....

.....

..... [2]

(b) Some of the planets in the Solar System have several moons (satellites) that have circular orbits about the planet.
The planet and each of its moons may be considered to be point masses.

Show that the radius x of the moon's orbit is related to the period T of the orbit by the expression

$$GM = \frac{4\pi^2 x^3}{T^2}$$

where G is the gravitational constant and M is the mass of the planet. Explain your working.

[3]

For Examiner's Use

(c) The planet Neptune has eight moons, each in a circular orbit of radius x and period T. The variation with T^2 of x^3 for some of the moons is shown in Fig. 2.1.

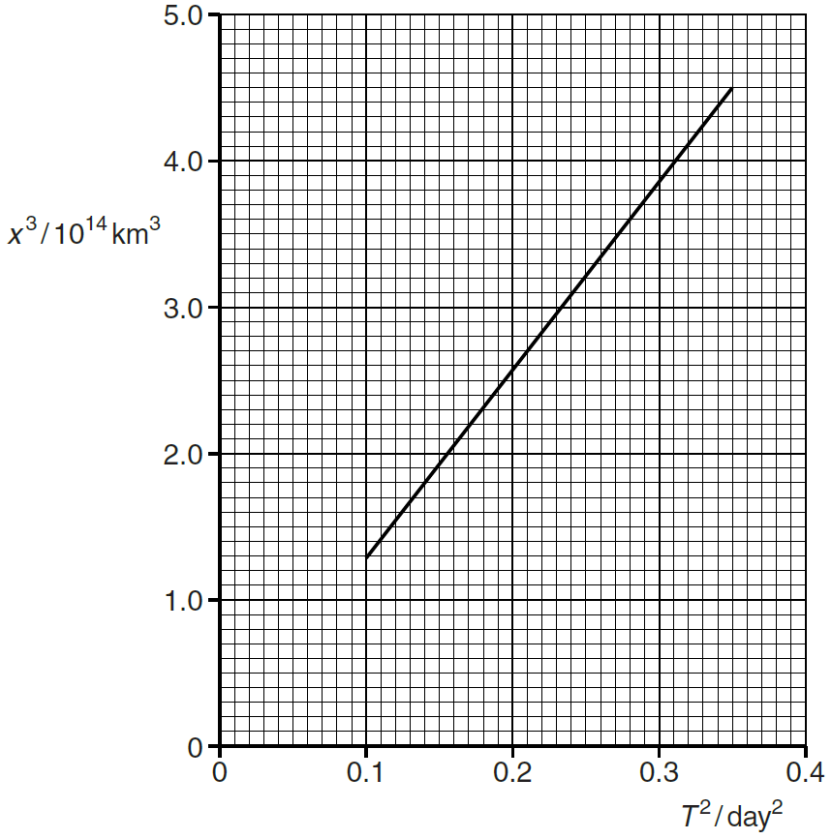

Fig. 2.1

(i) Show that the gradient of the line in Fig. 2.1 is $1.7 \times 10^{14}\ m^3\ s^{-2}$.

[1]

(ii) Hence determine the mass of Neptune.

mass = ... kg [2]

(d) The mass of the planet Uranus is smaller than that of the planet Neptune.
On Fig. 2.1, sketch the variation with T^2 of x^3 for the moons of Uranus.

[2]